Course Name: Computer Networks and Internet Protocol

Assignment - Week 10 (Jan 2026)

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10 **Total mark: 10 X 1 = 10**

QUESTION 1

Which is NOT one of the LLC services?

a) Connection mode service
b) Acknowledged connectionless service
c) Intermediate switching service
d) Unacknowledged connectionless service

Correct Answer: (c)

Detailed Solution: LLC layer provides three services: connection mode service, acknowledged connectionless service, and unacknowledged connectionless service. Intermediate switching service is not defined as an LLC service.

QUESTION 2

IEEE 802.1 is concerned with which issue in LANs and MANs?

a) Error handling
b) Networking
c) Internetworking
d) Flow control

Correct Answer: (c)

Detailed Solution: IEEE 802.1 deals with internetworking issues in LANs and MANs, such as bridging, VLANs, and network management.

QUESTION 3

Which sublayer of the data link layer performs data link functions that depend upon the type of medium?

a) Logical link control sublayer
b) Media access control sublayer
c) Network interface control sublayer
d) Error control sublayer

Correct Answer: (b)
Detailed Solution: The MAC sublayer of the data link layer handles functions that depend on the physical transmission medium, such as frame addressing and controlling access to the shared medium like CSMA/CD.

QUESTION 4

In 10Base-T and 100Base-T Ethernet technology, what does the letter "T" represent?

a) Transmission
b) Twisted Pair
c) Total utilization
d) Top Speed

Correct Answer: (b)
Detailed Solution: In 10Base-T and 100Base-T Ethernet, the "T" stands for Twisted Pair, meaning the network uses twisted-pair copper cables for data transmission.

QUESTION 5

In Go-Back-N ARQ, if the window size is 63, what is the range of sequence numbers?

a) 1 to 63
b) 1 to 64
c) 0 to 63
d) 0 to 64

Correct Answer: (c)
Detailed Solution: In Go-Back-N ARQ, if the window size is 2^m -1, the sequence numbers range from 0 to 2^m - 1. For window size 63, the sequence numbers are 0 to 63.

QUESTION 6

When 2 or more bits in a data unit have been changed during the transmission, the error is called ____________.

a) Random error
b) Burst error
c) Inverted error
d) Double error

Correct Answer : (b)

Detailed Solution: When a single bit error occurs in a data, it is called single bit error. When more than a single bit of data is corrupted or has an error, it is called burst error.

QUESTION 7

Which of the following are correct about Asynchronous MAC techniques?

I. Reservation method is suited for streaming voice traffic.
II. Contention method is suitable for continuous traffic.

a) Only I
b) Only II
c) Both I and II
d) Neither I nor II

Correct Answer: (a)

Detailed Solution: The reservation method works well for continuous or real time traffic, while contention methods are better suited for bursty traffic, not continuous traffic.

QUESTION 8

A pure ALOHA network transmits 200 bit frames on a shared channel with a data rate of 200 kbps. If the system (all stations together) generates 250 frames per second, what is the throughput of the network (in frames per second)?

a) 38 frames
b) 48 frames
c) 96 frames
d) 126 frames

Correct Answer: (a)

Detailed Solution: If the system creates 250 frames per second, or 1/4 frames per millisecond, then G = ¼. S = [G × e^(-2G)] = 0.152 ; Throughput = 250 × 0.152 = 38. Only 38 frames out of 250 will survive.

QUESTION 9

Which of the following is/are true?

I. In full-duplex switched Ethernet, there is no need for the CSMA/CD method.
II. Ethernet is best utilized when the load is heavy.
III. Slotted ALOHA has lower maximum utilization compared to pure ALOHA.
IV. Random backoff in CSMA/CD reduces the probability that stations retransmit at the same time after a collision.

a) I and IV
b) I and III
c) II and III
d) II and IV

Correct Answer: (a)
Detailed Solution: In full-duplex switched Ethernet, collisions do not occur, so CSMA/CD is not needed, making statement I true. Ethernet works best under light load, so II is false. Slotted ALOHA(≈36.8%) has higher utilization than pure ALOHA(≈18.4%), so III is false. Random backoff in CSMA/CD prevents repeated collisions, making IV true.

QUESTION 10

A network using CSMA/CD has a bandwidth of 10 Mbps. If the maximum propagation delay is 25.6 μs, what is the minimum frame size required to detect collisions?

a) 128 bytes
b) 32 bytes
c) 16 bytes
d) 64 bytes

Correct Answer: (d)
Detailed Solution: In CSMA/CD, the frame transmission time must be at least 2 × propagation delay = 51.2 μs. At 10 Mbps, the minimum frame size = 51.2 μs × 10 Mbps = 512 bits. This corresponds to 512 bits = 64 bytes.
